Exercise 3

To make exercise 1 completely linear, autonomous, and time-invariant, modify the expression of cardiac output as follows

$q = K^* P_{ra}$

that is, consider the offset term null. Then, recalculate the value of *K* to ensure the same equilibrium condition as in exercise 1.

1) Write the equations of the linear time-invariant autonomous system and determine the matrix *A*

$$\frac{dX(t)}{dt} = AX(t)$$

2) Using the Matlab command "eig," determine the eigenvalues and eigenvectors of *A*.

3) Write the free motion of the system as a function of the eigenvectors and eigenvalues. Imagining an initial perturbed state [Psa (1) Psv (1) Pra (1)] at choice, determine the constants, $\square_i$, of the solution and draw the graphs of the three state variables.

4) Write the free motion again by using the exponential function (Matlab command "expm").

5) Compare the graphs with those obtained numerically with Euler's method.

6) Vary the integration step of Euler's to find its effect and determine at which value a numerical instability occurs.

Model assumption. To make the cardiovascular model linear, autonomous, and time-invariant, the cardiac output is written as q = K·Pra, so the offset term is removed. The equilibrium pressures are kept equal to the previous exercise: Psa0 = 100 mmHg, Psv0 = 5 mmHg, Pra0 = 4 mmHg, and q0 = 5 L/min.

Computed parameters

Quantity	Expression	Value
q0	5/60 × 1000	83.333 ml/s
K	q0 / Pra0	20.833 ml/(s·mmHg)
Rsa	(Psa0 − Psv0) / q0	1.140 mmHg·s/ml
Rsv	(Psv0 − Pra0) / q0	0.0120 mmHg·s/ml
Csa	assigned	4 ml/mmHg
Csv	assigned	111 ml/mmHg
Cra	assigned	31 ml/mmHg

With state vector x(t) = [Psa(t) Psv(t) Pra(t)]^T and Ii(t) = 0, the system becomes:

dPsa/dt = (q − (Psa − Psv)/Rsa) / Csa

dPsv/dt = ((Psa − Psv)/Rsa − (Psv − Pra)/Rsv) / Csv

dPra/dt = ((Psv − Pra)/Rsv − q) / Cra

q = K·Pra

therefore x_dot = A x

A = [-0.219298 0.219298 5.208333
0.007903 -0.758653 0.750751
0.000000 2.688172 -3.360215] (s^-1)

The equilibrium condition is preserved because K is chosen as q0/Pra0. In fact, A·[100 5 4]^T = 0, so the assigned equilibrium remains an equilibrium of the new linear model.

2) Eigenvalues and eigenvectors. Using the MATLAB command eig(A), the eigenvalues are approximately:

lambda1 = 0

lambda2 = -0.360132 s^-1

lambda3 = -3.978034 s^-1

A corresponding set of eigenvectors, in the same order returned by MATLAB, is:

$v1 \approx [\ 0.997956 \quad 0.049898 \quad 0.039918\ ]^T$

$v2 \approx [\ -0.999252 \quad 0.028802 \quad 0.025807\ ]^T$

$v3 \approx [\ -0.800873 \quad -0.134133 \quad 0.583619\ ]^T$

Interpretation. The two negative eigenvalues correspond to decaying modes, while the zero eigenvalue indicates conservation of the total systemic filling volume. Therefore the model has a one-dimensional equilibrium manifold, not a single isolated equilibrium point.

3) Free motion from an initial perturbed state. Using the same perturbation as in the code, $x(0) = [150\ \ 5\ \ 4]^T$ mmHg, the modal form is:

$$x(t) = alpha1 \cdot v1 \cdot exp(lambda1\ t) + alpha2 \cdot v2 \cdot exp(lambda2\ t) + alpha3 \cdot v3 \cdot exp(lambda3\ t)$$

$$alpha = V^{(-1)}\ x(0) \approx [118.7784 \quad -31.5892 \quad 0.1265]^T$$

Because lambda3 is much more negative, its mode dies out quickly. The slow transient is mainly governed by lambda2. Since lambda1 = 0, the state does not return exactly to the original equilibrium $[100\ \ 5\ \ 4]^T$ unless the initial total filling volume is preserved. With $x(0) = [150\ \ 5\ \ 4]^T$, the total filling volume is larger than the nominal one, so the trajectories converge to another equilibrium on the same equilibrium line, approximately $[118.54\ \ 5.93\ \ 4.74]^T$ mmHg.

4) Free motion with the matrix exponential. The same free motion can be written compactly as:

$$x(t) = expm(A\ t)\ x(0)$$

This is exactly the formulation implemented in the script with XX(:,j) = expm(A*t(j))*X0. The result must coincide with the eigenvalue/eigenvector expression, because both represent the exact analytical solution of the same linear system.

5) Comparison with Euler's method. For a small integration step such as DT = 0.1 s, the Euler solution, the modal solution, and the matrix-exponential solution are practically superimposed. The numerical curves follow the exact solution very closely, so the three graphs should almost overlap. Any visible discrepancy is due only to discretization error.

6) Effect of the integration step and numerical instability. For explicit Euler applied to x_dot = A x, stability requires $|1 + DT \cdot lambda_i| < 1$ for the decaying eigenvalues. The most restrictive mode is lambda3 = -3.978034 s^-1, hence the critical step is approximately:

$$DT_crit \approx 2 / 3.978034 \approx 0.503\ s$$

Therefore, for DT below about 0.50 s the method remains stable. Around DT ≈ 0.25 s the fastest mode begins to alternate in sign, so damped numerical oscillations may appear. When DT exceeds about 0.503 s, the factor (1 + DT·lambda3) leaves the unit circle and the Euler solution becomes numerically unstable, even though the real physical system is still stable. This agrees with the code comment DT = 0.51, which is already slightly above the stability limit.

Conclusion. After removing the offset in the cardiac output law, the systemic circulation model becomes a linear autonomous LTI system. Its exact solution can be written either through the modal decomposition or through expm(A t). Euler integration reproduces the exact dynamics well for small time steps, but it becomes unstable for DT values larger than about 0.503 s.